

📄 Project Documentation – Global Electronics Retailer Analysis

0. Project Overview

Project Title: Global Electronics Retailer – Sales & Performance Analysis

Data Source: CSV files provided by the client (Sales, Products, Customers, Stores, Exchange Rates).

Objective:

Analyze global electronics sales performance (revenue, profit, margin, customers, products, regions) and build a Power BI dashboard for decision making.

Team Members:

- Yahiya Emad Eldin
- Mohamed Yahiya
- Mohamed Osama
- Ahmed Shaarawy
- Ahmed Abdel Hamid

1. Business Understanding

Lead: Yahiya Emad Eldin

Supporting: Ahmed Abdel Hamid

- Understand the business model of a global electronics retailer (multi-country, multi-currency, multiple product categories).
- Define key business questions:
- Are sales and profit trending up or down?
- Which countries, stores, and products drive most of the performance?
- Who are the most valuable customers?

Define main KPIs:

Total Sales, Total Profit, Margin %, Number of Orders, Units Sold.

Decide on main dashboard pages:

Executive, Product, Customer, Sales (Time & Region), Stores/Operations.

Outcome: Clear list of KPIs and dashboard pages aligned with business owner needs.

2. Data Understanding

Lead: Mohamed Yahiya

Supporting: Yahiya Emad Eldin

Files Analyzed:

- Sales.csv: Order-level transactions (Order Date, Delivery Date, ProductKey, CustomerKey, StoreKey, Quantity, Currency Code).
- Products.csv: Product details (Name, Brand, Category, Subcategory, Unit Price, Unit Cost).
- Customers.csv: Customer demographics (Location, Country, Birthday, Gender).
- Stores.csv: Store locations and opening dates.
- Exchange_Rates.csv: Daily exchange rates per currency vs USD.

Tasks:

- Check row counts, missing values, and basic distributions.
- Understand how each table connects:
 - Sales ↔ Products via ProductKey
 - Sales ↔ Customers via CustomerKey
 - Sales ↔ Stores via StoreKey
 - Sales ↔ Exchange_Rates via Order Date + Currency Code

Detect issues:

- Missing Delivery Dates
- Different currencies
- Date ranges across years
- Outcome: Clear mapping of all tables and data quality issues.

3. Data Preparation

Lead: Mohamed Osama

Supporting: Ahmed Shaarawy

Steps:

- Clean Dates
- Convert Order Date and Delivery Date to proper date format.
- Fix invalid/missing Delivery Date values by setting them as a few days after Order Date in a logical way.
- Payment Method Column
- Add a new column Payment Method in Sales with realistic distribution (e.g. Credit Card, Cash, Digital Wallet) to support payment analysis.
- Currency Standardization
- Use Exchange_Rates.csv to convert all sales into a unified currency (e.g. USD).
- Calculate Sales_USD for each row based on quantity, unit price, and exchange rate.
- Profit Calculation
- Join Sales with Products.

Compute:

- $\text{Sales_Amount} = \text{Quantity} \times \text{Unit Price}$
- $\text{Cost_Amount} = \text{Quantity} \times \text{Unit Cost}$
- $\text{Profit} = \text{Sales_Amount} - \text{Cost_Amount}$
- $\text{Margin \%} = \text{Profit} / \text{Sales_Amount}$
- Date Table
- Build a separate Date table (Calendar) to use in Power BI:
- Columns: Date, Year, Quarter, Month, MonthName, Weekday, WeekdayName, IsWeekend.
- Use it for all time-intelligence calculations (YTD, MTD, YoY, etc.).
- Outcome: Clean, enriched data model ready for Power BI.

4. Data Modeling

Lead: Ahmed Shaarawy

Supporting: Mohamed Osama

Model Design:

Fact Table:

- FactSales (from Sales.csv after preparation)
- Includes: Order Date, Delivery Date, ProductKey, StoreKey, CustomerKey, Payment Method, Sales_USD, Profit, Margin %, Quantity.
- Dimension Tables:
- DimDate (Date Table)
- DimProduct (from Products.csv)
- DimCustomer (from Customers.csv)
- DimStore (from Stores.csv)
- Optional: DimCurrency from Exchange_Rates if needed.

Relationships:

- FactSales[Order Date] → DimDate[Date]
- FactSales[ProductKey] → DimProduct[ProductKey]
- FactSales[CustomerKey] → DimCustomer[CustomerKey]
- FactSales[StoreKey] → DimStore[StoreKey]
- Outcome: Star schema optimized for reporting and DAX.

5. DAX Measures & Calculations

Lead: Ahmed Abdel Hamid

Supporting: Ahmed Shaarawy

- Core Measures (examples):
- Total Sales = SUM(FactSales[Sales_USD])
- Total Profit = SUM(FactSales[Profit])
- Margin % = DIVIDE([Total Profit], [Total Sales])
- Total Quantity = SUM(FactSales[Quantity])
- Time Intelligence:
- Sales MTD = TOTALMTD([Total Sales], DimDate[Date])
- Sales YTD = TOTALYTD([Total Sales], DimDate[Date])
- Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date]))
- Sales MTD LY = CALCULATE([Sales MTD], SAMEPERIODLASTYEAR(DimDate[Date]))
- Sales Var vs LY = [Total Sales] - [Sales LY]
- Sales Var % vs LY = DIVIDE([Sales Var vs LY], [Sales LY])

Executive Indicators:

- On-track/off-track flags based on comparison to target table (if used) or last year:
- Status = IF([Sales Var % vs LY] > 0, "On Track", "Off Track")
- KPI icons (▲ / ▼) using conditional formatting.
- Outcome: Set of reusable measures powering all dashboard visuals.

6. Visualization & Dashboard Design

Lead: Ahmed Abdel Hamid

Supporting: Yahiya Emad Eldin

- Page 1 – Executive Overview
- Cards: Total Sales, Total Profit, Margin %, Orders, Customers.
- Trend chart: Sales & Profit over time.
- KPIs: On Track / Off Track vs last year.
- Top 5 Products, Top 5 Countries, Top 5 Stores.

Page 2 – Product Performance

- Sales & Profit by Category, Subcategory, and Product.
- Top/Bottom products by margin.
- Product contribution to total sales.
- Filters: Category, Brand.

Page 3 – Customer Insights

- Sales by Customer Segment / Country.
- Top Customers by sales and profit.
- Customer distribution by age (if derived from Birthday).
- New vs Returning customers (if logic prepared).

Page 4 – Sales (Time & Region)

- Sales by Year, Quarter, Month.
- Seasonal trends.
- Map: Sales by Country/Region.
- Channel or Payment Method breakdown if needed.

Page 5 – Stores / Operations

- Store ranking by Sales, Profit, Margin.
- Store performance vs average.
- Open Date impact on performance (new vs old stores).
- Outcome: A professional Power BI report aligned with business and analyst needs.

7. Insights & Recommendations

Lead: Yahiya Emad Eldin

Supporting: Whole Team

- Identify high-performing countries and categories to focus marketing/supply.
- Detect low-margin products that may need price adjustment or cost control.
- Highlight key customers and regions for loyalty campaigns.
- Suggest monitoring of payment methods usage (e.g. high Credit Card usage, opportunity for installment offers).

8. Final Deliverables

Lead: Ahmed Abdel Hamid

- Power BI Report (.pbix) with 5 pages.
- Clean, enriched CSV files (Sales fact + dimension tables).
- Date Table CSV for calendar & time intelligence.
- This project documentation as a PDF/Word file.